

Hawkeye EQMS · Batch / Manufacturing Records · Feature Guide

Pharma-EQMS comparison · click-by-click guide · personas + lifecycle walkthrough · AI map · test results.

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1. Module overview

Module	Batch / Manufacturing Records
Pharma purpose	Capture the full manufacturing record for a single batch — BOM actuals, in-process tests, yield, equipment used, linked deviations — then drive QA review to final disposition (release / reject / quarantine).
Compliance basis	21 CFR 211.188 (batch production records) · 21 CFR 211.192 (production record review) · EU GMP Annex 11 (electronic batch records) · ICH Q7 §6.5
Hawkeye routes	/batch-records (frontend) · /api/batch-records (backend)
Hawkeye model file	backend/src/models/BatchRecordModel.js
Tenant module flag	modules.BATCH_RECORDS

Batches move MANUFACTURING → UNDER_REVIEW / PENDING_LAB_RESULTS / PENDING_QA_REVIEW → PENDING_DEVIATION_CLOSURE / PENDING_DISPOSITION → RELEASED / REJECTED / QUARANTINED. Operator captures BOM actuals + in-process tests + yields. QA reviews. VP disposes. Open deviations block release.

2. Standard pharma EQMS vs Hawkeye — gap analysis

Each row is a capability a pharma QA team would expect from an EQMS deviation module. Hawkeye column shows what's actually shipped. **MET** = parity, **PARTIAL** = ships but with caveats, **GAP** = not implemented yet.

Pharma expectation	Hawkeye implementation	Status	Note
Bill of Materials with theoretical vs actual qty 21 CFR 211.188(b)(2)	billOfMaterials[] subdocument with theoreticalQty + actualQty + supplierId + lot.	MET	
In-process tests with spec + result + tester + timestamp 21 CFR 211.110	inProcessTests[] subdocument: testName + spec + result (PASS/FAIL/null) + testedBy + testedAt + unit.	MET	
Yield tracking per stage with within-spec flag 21 CFR 211.103	yieldRecords[] subdocument with stage + percent + withinSpec.	MET	
Equipment used (linked to Equipment module) 21 CFR 211.188(b)(7)	equipmentUsed[] subdocument with equipmentId + start/end times.	MET	
Linked deviations gate release 21 CFR 211.192	linkedDeviationIds[] checked at /qa-review; if any not closed → status=PENDING_DEVIATION_CLOSURE.	MET	
QA review must complete before VP disposition 21 CFR 211.192	qa-review endpoint required before dispose. qaReviewedBy + qaReviewedAt + qaReviewNotes.	MET	
VP disposition: RELEASED / REJECTED / REWORK / REPROCESS / QUARANTINED 21 CFR 211.165	/dispose endpoint with decision enum.	MET	
Auto-set releaseDate on RELEASE decision 21 CFR 211.165(d)	releaseDate auto-set when disposition=RELEASED.	MET	
Lab results complete check before disposition 21 CFR 211.192	labResultsComplete boolean; if false → status=PENDING_LAB_RESULTS.	MET	
Stability sample assignment 21 CFR 211.166	Field exists; manual assignment.	PARTIAL	Auto-assign retained samples on release
21 CFR Part 11 e-signature on release 21 CFR Part 11	Generic e-sig endpoint exists; not enforced on /dispose.	PARTIAL	Wire e-sig requirement on RELEASE/REJECT

3. Personas + role mapping

Login password for every persona: `EqmsDemo@2026` . Roles drive which API endpoints respond — see Hawkeye `roleMiddleware.permit()` .

Persona	Email	Responsibilities in this module	Lifecycle states they touch
Michael Foster Production Head (admin · operator)	<code>production.head@novex-pharma.demo</code>	Creates batch record, captures BOM + tests + yields, submits for review.	MANUFACTURING UNDER_REVIEW

Persona	Email	Responsibilities in this module	Lifecycle states they touch
Aisha Patel QC Lab Lead (admin)	qc.lab@novex-pharma.demo	Completes lab analytical tests, marks labResultsComplete.	PENDING_LAB_RESULTS
James Thompson Head of QA (admin · reviewer)	qa.head@novex-pharma.demo	Reviews batch record, verifies deviation closure, advances to disposition.	PENDING_QA_REVIEW PENDING_DEVIATION_CLOSURE
Elena Vasquez VP Quality (tenant_admin · disposer)	vp.quality@novex-pharma.demo	Final disposition: RELEASE / REJECT / QUARANTINE.	PENDING_DISPOSITION RELEASED REJECTED QUARANTINED

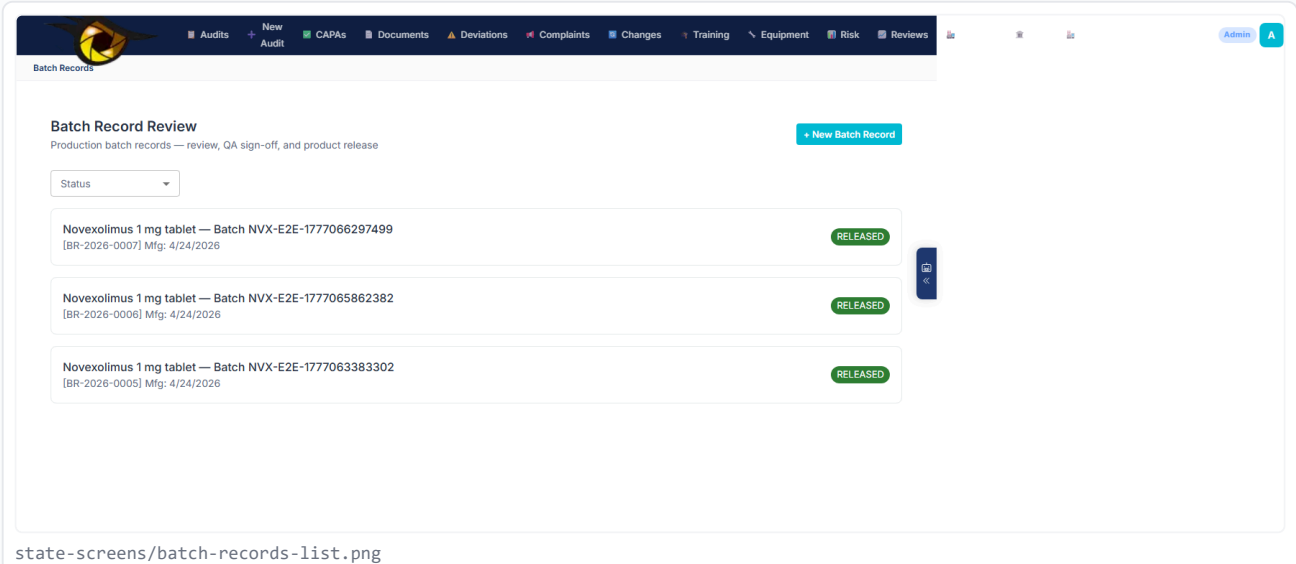
4. Feature catalogue (click-by-click)

4.1 Batch register

What it does	Lists all batches with status chip + product + batchNumber + manufacturingDate.
Menu / route	/batch-records
Available to roles	any tenant viewer
Backend API	GET /api/batch-records

CLICK-BY-CLICK

- NAVIGATE** Click 'Batches' in the top nav
→ Page renders
- WAIT** Spinner clears
→ Rows visible



4.2 + Create Batch dialog

What it does	Operator creates a new batch in MANUFACTURING state.
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Menu / route	/batch-records · top-right '+ Create Batch' button
Available to roles	operator · admin
Backend API	POST /api/batch-records

CLICK-BY-CLICK

- CLICK** Click '+ Create Batch'
 - Dialog opens
- TYPE** Fill batchNumber (e.g. NVX-2026-B015)
 - required
- TYPE** Fill productName
 - required
- CLICK** Pick manufacturingDate
 - required
- CLICK** + Add billOfMaterials rows (materialName + theoreticalQty + actualQty + unit)
 - Multiple rows
- CLICK** + Add inProcessTests rows (testName + spec + result + testedBy)
 - Multiple rows
- CLICK** Click 'Save'
 - Row appears with status=MANUFACTURING

FIELDS CAPTURED

Field	Required	Validation / values	Notes
batchNumber	Yes	string · user-defined	
productName	Yes	string	
manufacturingDate	Yes	ISO date	
billOfMaterials	No	[(materialName, theoreticalQty, actualQty, unit, supplierId, lot)]	
inProcessTests	No	[(testName, spec, result, testedBy, testedAt, unit)]	

4.3 Submit for review

What it does	Operator submits the batch for review. Routes to PENDING_LAB_RESULTS or PENDING_QA_REVIEW based on labResultsComplete flag.
Menu / route	Batch row · 'Submit for Review' button
Available to roles	operator
Backend API	POST /api/batch-records/:id/submit-for-review

CLICK-BY-CLICK

- CLICK** Click 'Submit for Review'
 - Drawer with labResultsComplete + labResultsSummary + linkedDeviationIds
- CLICK** Tick labResultsComplete=true (if QC done)
 - drives next state
- TYPE** Fill labResultsSummary
- CLICK** Click 'Submit'

→ Status flips to PENDING_QA_REVIEW (or PENDING_LAB_RESULTS if labResultsComplete=false)

4.4 QA review

What it does	QA reviews the batch record + verifies deviation closure + advances to disposition.
Menu / route	Batch row · 'QA Review' button (visible to QA role)
Available to roles	QA · admin
Backend API	POST /api/batch-records/:id/qa-review

CLICK-BY-CLICK

1. **CLICK** Click 'QA Review'
→ Drawer with qaReviewNotes + deviationsResolved + labResultsComplete checks
2. **TYPE** Fill qaReviewNotes
→ required
3. **CLICK** Tick deviationsResolved + labResultsComplete (both must be true)
→ if either false → status=PENDING_*
4. **CLICK** Click 'Submit'
→ Status flips to PENDING_DISPOSITION (when both checks pass)

4.5 Dispose (VP)

What it does	Final disposition. RELEASED auto-sets releaseDate.
Menu / route	Batch row · 'Dispose' button (visible to VP / tenant_admin)
Available to roles	VP · tenant_admin
Backend API	POST /api/batch-records/:id/dispose

CLICK-BY-CLICK

1. **CLICK** Click 'Dispose'
→ Drawer with decision picker
2. **CLICK** Pick decision (RELEASED / REJECTED / REWORK / REPROCESS / QUARANTINED)
→ required
3. **TYPE** Fill justification
→ required
4. **CLICK** (Optional) e-sig
5. **CLICK** Click 'Confirm + Sign'
→ Status flips; releaseDate auto-set if RELEASED

5. Full lifecycle walkthrough — one transaction, end-to-end

One batch walked from manufacturing through QA review to RELEASE. Personas: Michael (operator) → James (QA) → Elena (VP).

#	Persona	State transition	Action	API	Observed	Result
1	Michael Production Head	— → MANUFACTURING	+ Create Batch → batchNumber NVX-E2E-NNN / Novexolimus 1mg / today / BOM (Novexolimus API + Lactose) / inProcessTests (Blend uniformity PASS, Compression force PASS)	POST /api/batch-records	Row visible · status=MANUFACTURING	PASS

Expected DB state after this step:

```
batch-records { _id, batchNumber, productName, manufacturingDate, billOfMaterials: [...], inProcessTests: [...], status: 'MANUFACTURING', createdBy: michael._id }
```

2	Michael Operator	MANUFACTURING → PENDING_QA_REVIEW	Submit for review · labResultsComplete=true · linkedDeviationIds=[]	POST /api/batch-records/:id/submit-for-review	Status=PENDING_QA_REVIEW (lab complete + no deviations)	PASS
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Expected DB state after this step:

```
batch-records { status: 'PENDING_QA_REVIEW', labResultsComplete: true, labResultsSummary }
```

3	James Head of QA	PENDING_QA_REVIEW → PENDING_DISPOSITION	QA Review · qaReviewNotes · deviationsResolved=true · labResultsComplete=true	POST /api/batch-records/:id/qa-review	Status=PENDING_DISPOSITION	PASS
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Expected DB state after this step:

```
batch-records { status: 'PENDING_DISPOSITION', qaReviewedBy: james._id, qaReviewedAt, qaReviewNotes }
```

4	Elena VP Quality	PENDING_DISPOSITION → RELEASED	Dispose · decision=RELEASED · justification 'all release criteria met per SOP-QC-014'	POST /api/batch-records/:id/dispose	Status=RELEASED · releaseDate auto-set	PASS
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#	Persona	State transition	Action	API	Observed	Result
Expected DB state after this step:						
<pre>batch-records { status: 'RELEASED', disposition: { decision: 'RELEASED', justification, decidedBy: elena._id }, releaseDate: <now> }</pre>						

6. AI assistance map

Each AI assist is a separate endpoint. Outputs flow through the grounded-generation runtime (citation gate, confidence floor, schema validation, accept/reject audit trail).

AI agent	Attached to state(s)	Endpoint	Where in UI	What it returns	Provider
(roadmap) Yield anomaly detector (Wave 3)	MANUFACTURING UNDER_REVIEW	(future)	(future)	Compare current yield + in-process trends vs historical → flag outliers for QA	Statistical (no LLM)
(roadmap) AI batch-record review assistant	PENDING_QA_REVIEW	(future)	(future)	AI scans the BR for missing entries / out-of-spec results / spec-drift across batches	Free Gemini

7. Regulator-trace matrix

For each lifecycle state, the specific clauses + evidence Hawkeye captures so an inspector can audit the record. Maps to FDA / ISO / ICH / EU GMP source clauses.

State	Cited clauses	Evidence captured	Backing records
MANUFACTURING	21 CFR 211.188(b)	batchNumber + productName + manufacturingDate + BOM + inProcessTests + equipmentUsed	batch-records
PENDING_QA_REVIEW	21 CFR 211.192	labResultsComplete=true + labResultsSummary	batch-records
PENDING_DISPOSITION	21 CFR 211.192	qaReviewedBy + qaReviewedAt + qaReviewNotes + deviationsResolved=true	batch-records
RELEASED	21 CFR 211.165(d) 21 CFR Part 11 §11.50	disposition.decision + justification + decidedBy + releaseDate + (e-sig)	batch-records + (electronic-signatures)
REJECTED	21 CFR 211.165(f)	disposition.decision + justification	batch-records

8. Test results matrix

Suite	Scope	Result	Evidence
eqms-lifecycle.spec.ts · batch-record	MANUFACTURING → PENDING_QA_REVIEW → PENDING_DISPOSITION → RELEASED	PASS	5/5 PASS · eqms-test-results-v2.pdf

9. Known gaps + roadmap

- **Mandatory e-sig on RELEASE / REJECT** — Wire /api/electronic-signatures/sign on /dispose. Captures meaning='RELEASED' or 'REJECTED'.
HIGH
- **Stability sample auto-assignment on RELEASE** — Auto-create retained sample record + scheduling per stability protocol. **HIGH**
- **AI yield anomaly detector** — Wave 3 statistical detector flags yields outside $\pm 2\sigma$ of last 30 batches. **MEDIUM**
- **Cross-link to deviation drawer** — linkedDeviationIds clickable chips on batch detail. **MEDIUM**